MIS 522 - Practicum

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Assignment

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Group Questions - 1

1) Why is Pandas useful for MIS data analysis?

→ Pandas is useful because it makes it easy to load, inspect, clean, transform, merge, and analyze structured data. In MIS, analysts often work with large tables containing missing values, duplicates, invalid records, and inconsistent categories. Pandas provide tools such as Dataframes, filtering, grouping, merging and summary statistics to turn raw data into useful management information.

2) Why do we use Numpy?

→ Numpy is used for fast numerical calculations and data processing. It provides efficient arrays and mathematical functions that are useful for calculating values such as averages, totals, standard deviations, and other statistical measures. Pandas also relies heavily on Numpy for numerical operations.

3) Why are visualizations important for managers?

→ Visualizations help managers understand patterns, trends, comparisons, and unusual values quickly. For example, a chart can show which household categories have the highest economic burden or how expenses vary over time more clearly than a large table of numbers. This supports faster and better decision-making.

4) Why is reproducibility important when creating a teaching dataset?

→ Reproducibility means that the same dataset and analysis process can be re-created and produce consistent results. This is important in teaching because students can verify their work, instructors can compare results fairly, and errors can be identified more easily. In business analytics, reproducibility also improves trust, transparency, and auditability of management reports.

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Group Questions - 2

1) Why are there more than 3000 rows?

-> There are 3025 rows because the dataset contains the original 3000 household record plus 25 intentionally duplicates were added to stimulate a real-world data-quality problem. If they are not removed, they can cause household counts, income totals, and other statistics to be overstated.

2) Which variables contain NULL values?

-> The variables containing NULL values were:

Monthly-Income - 80 missing values were intentionally inserted.

Employment_Status - 55 missing values.

Education - 35 missing values.

Missing values can affect statistical analysis and may require appropriate treatment, such as imputation or removal, depending on the business purpose.

3) Which values are logically invalid?

-> The logically invalid values include:

Negative Monthly-Income - 20 records.

Negative Household_Size - 12 records.

Household size cannot logically be negative, and household income should not be represented as a negative amount in this particular dataset.

There are also inconsistent category values, such as "urban", "URBAN", and "rural", which need to be standardized.

4) Why is negative income a data-quality problem?

-> Negative income is problematic because the dataset is intended to represent monthly household earnings. A negative value could incorrectly reduce the calculated mean income, total income, and other economic indicators. Consequently, managers could receive a misleading picture of household economic conditions.

5) Why can inconsistent province spelling affect reports and dashboards?

-> Inconsistent spellings can cause the same province to be treated as different categories. For example, "Dogmati"

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,"bagmati", ~~Bagma~~ "BAGMATI", and "Bagmati Province" could appear as separate provinces in a report.

This can lead to incorrect province-level counts, averages, charts, and ~~names~~ comparisons. Standardizing category names is therefore essential for accurate management reporting and decision-making.

Group Question - 3

1) Why should expenditure normally not be negative?

-> Expenditure represents the amount of money spent by a household on goods and services. Therefore, it should normally be zero or positive. Negative values may indicate data-entry errors, incorrect calculations or refunds.

2) What is an orphan Household-ID?

-> An orphan Household-ID is an ID that doesnot match any household record in the main dataset. It may occur due to missing records, incorrect data entry, or mismatched IDs. These records should be checked for accuracy.

3) Why can duplicate monthly observations distort analysis?

-> Duplicate monthly observations occur when the same household has multiple records for the same month. They can cause expenditure to be counted more than once, making totals and averages inaccurate and leading to misleading results.

4) Which variables are time-series variables?

-> Time-series variables are recorded over different periods to observe changes over time. In our dataset, Observation_Month, Inflation_Rate, Food_Inflation, Interest_Rate, Essential_Expenditure, Discretionary_Expenditure, and Employment_Index can be analyzed over time.

5) Why should extreme expenditure values be investigated?

-> Extreme expenditure values are unusually high or low compared with other observations. They may result from data-entry errors, incorrect calculations, or genuine unusual spending. Investigating these values helps ensure accurate analysis & prevents misleading results.

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Group Question-4

1) Audit Table

Problem	Variable	Count	Treatment
Missing values	Employment_Status	55	Impute or mark Unknown
Missing values	Education	35	Impute or mark Unknown
Missing values	Monthly_Income	81	Investigate and impute
Duplicates	All variables	25	Verify & remove duplicates
Negative values	Monthly_Income	20	Verify and correct
Negative values	Household_size	12	Verify and correct
Potential outliers	Monthly_Income	To investigate	Check extreme incomes

2) Which problem should be fixed first?

→ Invalid values and duplicate records should be fixed first because they can lead to incorrect analysis. Next, handle missing values and investigate potential outliers.

3) Which cleaning decisions require business knowledge?

→ • Negative income: Determine whether it is an error or a legitimate financial situation.

• Missing values: Decide whether to impute values or mark them as "Unknown".

~~• Duplicates: Confirm whether unusually high incomes are genuine or errors.~~

• Income outliers: Check whether unusally high incomes are genuine or errors.

• Duplicates: Confirm whether records are truly duplicated.

• Household size: Define valid household - size limits and correct invalid values.

4) Why should the dirty raw data be preserved?

→ Raw data should be preserved for auditing, verification, and reproducibility. It also allows original values to be recovered if cleaning decisions are incorrect. Therefore,

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cleaning should be performed on a separate copy of the data.

Group Question-5

1) Which columns have missing values?

-> • Inflation_Rate : 120
• Essential_Expenditure: 100
• Discretionary_Expenditure: 70

2) Which columns contain negative values?

-> • Inflation_Rate: 7
• Interest_Rate : 8
• Essential_Expenditure: 10
• Discretionary_Expenditure: 8

3) Which negative values are invalid for this case?

-> Negative essential and discretionary expenditure values are likely invalid because expenditure is normally non-negative. Negative inflation and interest rates may be valid in certain economic situations, so they require investigation.

4) Which variables should be checked for outliers?

-> • Essential_Expenditure
• Discretionary_Expenditure
• Inflation_Rate
• Interest_Rate
• Employment_Index

For example, the maximum essential of 369,928 is unusually high and should be investigated.

5) What could happen if this dirty data were loaded directly into Power BI?

-> • Missing values could cause incomplete reports & misleading averages.
• Invalid negative expenditure values could distort calculations.

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- Duplicate records could inflate totals.
- Outliers could skew charts and summary statistics.
- Incorrect data could lead to misleading dashboards & poor business decisions.

Group Question – 6

1) Is the cleaned master table exactly 3000 households?
→ Yes, The cleaned table contains 3000 rows & 3000 unique household IDs.

2) Why was negative income converted to missing before imputation?
→ Negative income may be invalid in this dataset. Converting it to missing allows it to be replaced with a reasonable value instead of affecting calculations.

3) Why use median for income?
→ The median is less affected by extremely high or low income values than the mean, making it suitable for income data.

4) Why use mode for categorial variables?
→ The mode replaces missing values with the most frequently occuring category - preserving valid categorial values.

5) Why must Household_ID be unique in the master table?
→ Each household should have only master record. Unique ID's prevent duplicated households from distorting analysis and help link the master table with other datasets accurately.

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Group Questions-7

1) How many orphan observations were removed?
→ A total of 30 orphan observations were removed because their Household_ID did not exist in the cleaned master table.

2) Why should orphan records be removed before merging?
→ Orphan records cannot be matched to a valid household. Removing them prevents incorrect relationships and unreliable analysis.

3) Why flag outliers instead of automatically deleting them?
→ Outliers may represent genuine high expenditures or data-entry errors. Flagging them allows further investigation without losing potentially useful data.

4) What is the danger of replacing extreme values with the mean?
→ The mean can distort the original data distribution and hide important differences. It may also be strongly influenced by extreme values, leading to misleading analysis.

Group Questions-8

1) Is income skewed?
→ Yes, monthly income is right-skewed, as most households earn moderate incomes while a few earn very high incomes.

2) Is expenditure skewed?
→ Yes, essential expenditure is strongly right-skewed, with most values concentrated at the lower end and a few extremely high values.

3) Which variables has stronger evidence of extreme observations?
→ Essential expenditure shows stronger evidence of extreme observations because a few unusually high values extend far beyond the main distribution.

4) Why is median useful for skewed economic data?
→ The median is less affected by extreme values than the mean, making it a more reliable measure of typical household income and expenditure.

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Group Question-9

1) Which province has the largest sample?
-> Bagmati has the largest sample, with 789 households.

2) Which employment category is largest?
-> Employed is the largest category, with 1,313 households.

3) Is the dataset balanced across provinces?
-> No. Household counts vary across provinces, from 789 in Bagmati to 196 in Karnali.

4) Why can unequal group sizes affect interpretation?
-> Unequal group sizes can make comparisons less reliable because larger groups may dominate overall results while smaller groups may be less representative.

Group Question-10

1) Compare mean and median income.
-> • Mean income: 74,077.07
• Median income: 66,333.00

The mean is higher than the median.

2) What does the difference tell you?
-> It indicates that income is right-skewed, with a few high-income households raising the average.

3) Which variables has the greatest standard deviation?
-> Essential_Expenditure has the greatest standard deviation among the expenditure and economic indicating substantial variation in household spending.

4) Which variables are suitable for mode analysis?
-> Categorical variables such as Province, Employment_Status, Education, and Area_Type are suitable for mode analysis. Mode can also be used for numerical variables when repeated values are meaningful.

5) Why should managers not rely only on average values?
-> Averages can hide differences between households and be affected by extreme values. Managers should also examine the median, standard deviation, and distribution to make better-informed decisions.

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Group Question - 11

1) Which province has the highest average income?
-> Sudurpashchim, with an average income of 76,462.22.

2) Which has the highest median income?
-> Bagmati, with a median income of 66,422.00.

3) Are the rankings identical?
-> No, Sudurpashchim has the highest average income, while Bagmati has the highest median income.

4) Which provinces has the largest average household size?
-> Lumbini, with an average household size of 4.59 people.

5) What management decision could use this summary?
-> Managers can use this summary to allocate resources plan household support programs, and design income-assistance policies according to the needs of each province.

Group Question - 12

1) Why is the merged table larger than 3000?
-> The merged table contains 7,970 rows because Table 2 has 7,970 monthly observations for households while Table 1 contains household-level information.

2) Explain the one-to-many relationship.
-> One household in Table 1 can have multiple monthly observations in Table 2 because the same household is recorded across different months.

3) Why is many-to-one appropriate?
-> It is appropriate because many rows in Table 2 can match one household record in Table 1 through Household_ID.

4) What could happen if household_ID were duplicated in Table 1?
-> Duplicate IDs could create extra rows during merging, causing incorrect results and duplicated household information.

5) Why is merge validation important in MIS?
-> Merge validation ensures that datasets are combined correctly and helps prevent duplicate records, data errors, and misleading analysis.

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Group Question - 13

Choose Five engineered variable

1) Total Expenditure
Formula: Essential Expenditure + Discretionary Expenditure.
Measures: The total amount spend by a household.
Value: Shows overall household spending.
Management use: Helps prepare household budgets.

2) Income Per Person
Formula: Monthly Income ÷ Household Size.
Measures: Average income available per household member.
Value: Helps compare households of different sizes.
Management use: Helps identify households needing financial support.

3) Saving Estimate
Formula: Monthly Income − Total Expenditure.
Measures: The estimated money remaining after expenses.
Value: Shows whether a household has a surplus or deficit.
Management use: Helps with savings and financial planning.

4) Financial burden
Formula: Total Expenditure ÷ Monthly Income.
Measures: The proportion of income spent on expenses.
Value: Shows how much income is used for spending.
Management use: Helps identify households facing financial pressure.

5) Saving Rate
Formula: Saving Estimate ÷ Monthly Income
Measures: The proportion of income remaining after expenditure.
Value: Helps compare saving capacity across households.
Management use: Supports financial planning and savings strategies.

Key discussion: Why can two households with the same income have very different economic conditions?

→ Two household with the same income may have different household sizes, living costs, and spending habits. One household may save more, while another may struggle

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to cover essential expenses. Therefore, income alone cannot show the complete financial condition of a household.

Group Question - 14

1) Which income group spends the most?
-> The low-income group spends the most on average, with expenditure of NPR. 37,378.96.

2) Which has the highest financial burden?
-> The low-income group has the highest financial burden, at 0.99 or 99% of its income.

3) Does higher income automatically mean lower burden?
-> In this dataset, higher-income groups have a lower financial burden because their average expenditure remains almost the same while their incomes increases.

4) What might explain a high burden?
-> A high burden may result from low income, high living costs, large household size, or unexpected expenses.

5) Which group would you monitor if you were a bank/financial manager?
-> I would monitor the low-income group because it spends almost all its income, leaving little money for savings & emergencies.

Group Question - 15

1) What trend do you observe?
-> Inflation remains relatively stable, while food inflation & interest rates fluctuate slightly over time.

2) Does food inflation behave differently from overall inflation?
-> Yes, food inflation shows greater fluctuations than overall inflation, indicating more changes in food prices.

3) Why should managers monitor inflation and interest rates together?
-> Manager should monitor both because inflation affects operating costs, while interest rates affect borrowing costs and investment decisions.

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4) What happens if expenditure rises faster than income?
-> Household savings decrease, and financial burden increases, making it harder to manage daily expenses.

5) What additional economic variable would improve the analysis?
-> The unemployment rate would improve the analysis by showing how changes in employment affect household income & spending.

Group Questions - 16

1) Identify two strong relationships.
-> Monthly Income & Savings Estimate have a strong positive relationship (0.92), while Savings Estimate & financial Burden have a strong negative relationship (-0.79).

2) Identify one weak relationship.
-> Inflation Rate and Household size have a very weak relationship (0.00).

3) Which relationship is most useful for management?
-> The relationship between Monthly Income & Savings Estimate is useful for financial planning & assessing household savings capacity.

4) Why does correlation not prove causation?
-> Correlation shows that two variables are related, but it doesnot prove that one variable directly causes changes in the other.

5) What other variables could explain the observed relationships?
-> Household size, employment status, education level, living costs and unexpected expenses could help explain these relationships.

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Group Question - 17

1) Which income group shows greater expenditure variation?

→ The low-income group appears to show greater variation in expenditure, with a wider spread of values and several extreme outliers.

2) What does the boxplot show that a mean cannot?

→ A boxplot shows the median, spread, variation, and outliers, while the mean provides only the average value.

3) Should all outliers be deleted?

→ No, outliers should not all be deleted because some may represent genuine household spending. They should be investigated before making corrections.

4) Give one legitimate economic reason for unusually high expenditure.

→ A household may have unusually high expenditure due to medical emergencies, education fees, or major purchases.

Group Question - 18

Household Income Distribution Across Provinces

1) Which province has the highest median household income?

→ Bagmati Province appears to have the highest median household income, while Madesh and Gandaki also show relatively high incomes.

2) Which province shows the greatest income variation?

→ Madesh Province shows relatively wide income variation, while Bagmati has several extreme high-income values.

3) Are there any extreme income values?

→ Yes, all provinces have high-income outliers. Some households earn more than NPR 200,000 per month, with one Bagmati outlier exceeding NPR 300,000.

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4) Why is a boxplot useful for this analysis?
-> A boxplot helps compare median income, income variation and extreme values across provinces.

5) Why might average income alone be misleading?
-> Average income can be affected by a few very high-income households. The median often better represents the typical household's income.

6) What economic factors could explain difference between provinces?
-> Differences may result from employment opportunities, industrial development, education, urbanization, infrastructure, agricultural productivity and remittances.

Group Question -19

1) Which area has higher average income?
-> Rural area have a higher average income of Rs 74,347.26 compared to urban areas, which have Rs. 73,281.83.

2) Which area has higher average expenditure?
-> Rural areas have a higher average expenditure of Rs. 37,545.29, while urban areas have Rs. 37,217.28.

3) Which area has higher average savings?
-> Rural areas have higher average savings of Rs. 36,801.97 compared to urban areas, which have Rs. 36,064.55.

4) Which area has greater financial burden?
-> Urban areas have a slightly greater financial burden, with an average of 0.64 compared to 0.63 in rural areas.

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5) What economic factors might explain urban-rural difference?

→ Urban-rural differences may be explained by variations in the cost of living, employment opportunities, income sources, housing costs, transportation expenses, and access to services. Urban residents may face higher living costs, resulting in a greater financial burden, while rural households may have lower living expenses that allow them to save slightly more.

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